

1. Executive Summary

		Traffic Summary		Count	
Metric	Value	Total Flows Observed		35695	
Total Threats Mitigated	38	Total Packets Analyzed		2035405226	
High Priority	10	Malicious Packets		2033035217	
Low Priority	28	True Negatives Passed		23193	
Manual Releases	0				
Manual Blocks	0	Attack Vector		Count	
		ICMP Flood		6	
ML Metrics	Value	SYN Flood		14	
False Positives	6	UDP Flood		7	
FP Rate	0.03%	Uncertain		11	
IF Precision	99.95%				
IF Recall	93.59%	Action		Count	
IF F1-Score	96.67%	Quarantined		10	
IF Accuracy	97.65%	Time Ban		10	
		Blackhole		0	
		Released		3	

2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	99.95%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	93.59%	Share of actual attacks successfully flagged
F1-Score	96.67%	Balanced measure combining Precision and Recall
Accuracy	97.65%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.03%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	6.41%	Actual attacks that went undetected
True Positive Rate (TPR)	93.59%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.97%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 11963	FN 819
Actual: Normal	FP 6	TN 22374

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	100.00%	Share of classified flows assigned the correct attack type
Recall	100.00%	Share of attacks of each type that were correctly identified
F1-Score	100.00%	Balanced measure combining Precision and Recall
Accuracy	100.00%	Overall proportion of correct classifications

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	6716	0	0
Act: ICMP	0	1644	0
Act: UDP	0	0	3603

2c. Response Latency

Metric	Value	Description
Detection Time	491.0 ms	Average time taken to detect an attack
Mitigation Response Time	6.7 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2d. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	2.03%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	39.98%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	20.59%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.17	4	2	2026-09-15 09:35:54	2026-09-15 09:40:28
10.0.0.16	2	2	2026-09-15 09:39:10	2026-09-15 09:40:22
10.0.0.18	2	2	2026-09-15 09:39:11	2026-09-15 09:40:23
10.0.0.19	2	2	2026-09-15 09:39:16	2026-09-15 09:40:28
10.0.0.21	1	1	2026-09-15 09:39:42	2026-09-15 09:39:52
10.0.0.22	1	1	2026-09-15 09:39:43	2026-09-15 09:39:53
10.0.0.23	1	1	2026-09-15 09:40:13	2026-09-15 09:40:23
10.0.0.24	1	1	2026-09-15 09:40:12	2026-09-15 09:40:22
10.0.0.25	1	1	2026-09-15 09:40:15	2026-09-15 09:40:25
10.0.0.1	1	1	2026-09-15 09:40:01	2026-09-15 09:40:12
10.0.0.8	1	1	2026-09-15 09:40:00	2026-09-15 09:40:11

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-15 09:33:09	10.0.0.16	DDoS	Uncertain	-	Low	Quarantined
2026-09-15 09:33:10	10.0.0.18	DDoS	Uncertain	-	Low	Quarantined
2026-09-15 09:33:16	10.0.0.17	DDoS	Uncertain	-	Low	Quarantined
2026-09-15 09:33:19	10.0.0.16	DDoS	SYN Flood	93.2%	Critical	Time Ban (1m)
2026-09-15 09:33:20	10.0.0.16	DDoS	SYN Flood	93.2%	Critical	Time Ban
2026-09-15 09:33:20	10.0.0.19	DDoS	SYN Flood	89.5%	Critical	Quarantined
2026-09-15 09:33:21	10.0.0.18	DDoS	SYN Flood	91.2%	Critical	Time Ban (1m)
2026-09-15 09:33:22	10.0.0.18	DDoS	SYN Flood	91.2%	Critical	Time Ban
2026-09-15 09:33:27	10.0.0.17	DDoS	SYN Flood	93.0%	Critical	Time Ban (1m)
2026-09-15 09:33:27	10.0.0.17	DDoS	SYN Flood	93.0%	Critical	Time Ban
2026-09-15 09:33:31	10.0.0.19	DDoS	SYN Flood	90.9%	Critical	Time Ban (1m)
2026-09-15 09:33:31	10.0.0.19	DDoS	SYN Flood	90.9%	Critical	Time Ban
2026-09-15 09:33:40	10.0.0.20	DDoS	Uncertain	-	Low	Quarantined
2026-09-15 09:33:40	10.0.0.21	DDoS	Uncertain	-	Low	Quarantined
2026-09-15 09:33:43	10.0.0.22	DDoS	Uncertain	-	Low	Quarantined
2026-09-15 09:33:50	10.0.0.20	DDoS	UDP Flood	79.1%	High	Time Ban (1m)
2026-09-15 09:33:50	10.0.0.20	DDoS	UDP Flood	79.1%	High	Time Ban
2026-09-15 09:33:51	10.0.0.21	DDoS	UDP Flood	77.9%	High	Time Ban (1m)
2026-09-15 09:33:51	10.0.0.21	DDoS	UDP Flood	77.9%	High	Time Ban
2026-09-15 09:33:53	10.0.0.22	DDoS	UDP Flood	77.9%	High	Time Ban (1m)
2026-09-15 09:33:54	10.0.0.22	DDoS	UDP Flood	77.9%	High	Time Ban
2026-09-15 09:34:10	10.0.0.24	DDoS	Uncertain	-	Low	Quarantined
2026-09-15 09:34:10	10.0.0.23	DDoS	Uncertain	-	Low	Quarantined
2026-09-15 09:34:12	10.0.0.25	DDoS	Uncertain	-	Low	Quarantined
2026-09-15 09:34:20	10.0.0.16	DDoS	SYN Flood	93.2%	Critical	Time Ban (2m)
2026-09-15 09:34:20	10.0.0.24	DDoS	ICMP Flood	90.9%	Critical	Time Ban (1m)
2026-09-15 09:34:20	10.0.0.23	DDoS	ICMP Flood	90.7%	Critical	Time Ban (1m)
2026-09-15 09:34:21	10.0.0.23	DDoS	ICMP Flood	90.7%	High	Time Ban
2026-09-15 09:34:21	10.0.0.24	DDoS	ICMP Flood	90.9%	Critical	Time Ban
2026-09-15 09:34:21	10.0.0.18	DDoS	SYN Flood	91.2%	High	Time Ban (2m)
2026-09-15 09:34:22	10.0.0.25	DDoS	ICMP Flood	91.2%	Critical	Time Ban (1m)
2026-09-15 09:34:23	10.0.0.25	DDoS	ICMP Flood	91.2%	Critical	Time Ban
2026-09-15 09:34:28	10.0.0.17	DDoS	SYN Flood	93.0%	Critical	Time Ban (2m)
2026-09-15 09:34:32	10.0.0.19	DDoS	SYN Flood	91.6%	Critical	Time Ban (2m)
2026-09-15 09:36:05	10.0.0.17	DDoS	SYN Flood	92.7%	High	Released
2026-09-15 09:40:00	10.0.0.6	DDoS	UDP Flood	92.4%	High	Time Ban (1m)
2026-09-15 09:40:11	10.0.0.8	DDoS	Uncertain	68.4%	Low	Released

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-15 09:40:12	10.0.0.1	DDoS	Uncertain	60.1%	Low	Released

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.17	SYN Flood	0.6337	93.2%	Critical	Phase 2	2	1m 12s	2026-09-15 09:40:28	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6450	93.7%	Critical	Phase 2	2	1m 12s	2026-09-15 09:40:28	Attack persisted - escalating to time ban
10.0.0.25	ICMP Flood	0.6456	91.2%	High	Phase 2	1	10s	2026-09-15 09:40:25	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6384	94.2%	Critical	Phase 2	2	1m 12s	2026-09-15 09:40:23	Attack persisted - escalating to time ban
10.0.0.23	ICMP Flood	0.6445	90.9%	High	Phase 2	1	10s	2026-09-15 09:40:23	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6015	92.9%	High	Phase 2	2	1m 12s	2026-09-15 09:40:22	Attack persisted - escalating to time ban
10.0.0.24	ICMP Flood	0.6453	91.0%	High	Phase 2	1	10s	2026-09-15 09:40:22	Attack persisted - escalating to time ban
10.0.0.1	Uncertain	0.5924	60.1%	Low	Phase 1	1	11s	2026-09-15 09:40:12	Attack Stopped
10.0.0.8	Uncertain	0.6298	68.4%	Low	Phase 1	1	11s	2026-09-15 09:40:11	Attack Stopped
10.0.0.22	UDP Flood	0.6359	82.3%	High	Phase 2	1	10s	2026-09-15 09:39:53	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6458	78.2%	High	Phase 2	1	10s	2026-09-15 09:39:52	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6423	93.2%	Critical	Phase 2	1	11s	2026-09-15 09:39:27	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6516	93.7%	Critical	Phase 2	1	11s	2026-09-15 09:39:27	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6428	94.2%	Critical	Phase 2	1	11s	2026-09-15 09:39:22	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6385	92.9%	Critical	Phase 2	1	11s	2026-09-15 09:39:21	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6253	92.7%	Critical	Phase 2	1	1m 1s	2026-09-15 09:37:07	Ban Expired
10.0.0.17	SYN Flood	0.6058	92.7%	High	Phase 1	1	11s	2026-09-15 09:36:05	Attack Stopped

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME
Network Administrator

DATE